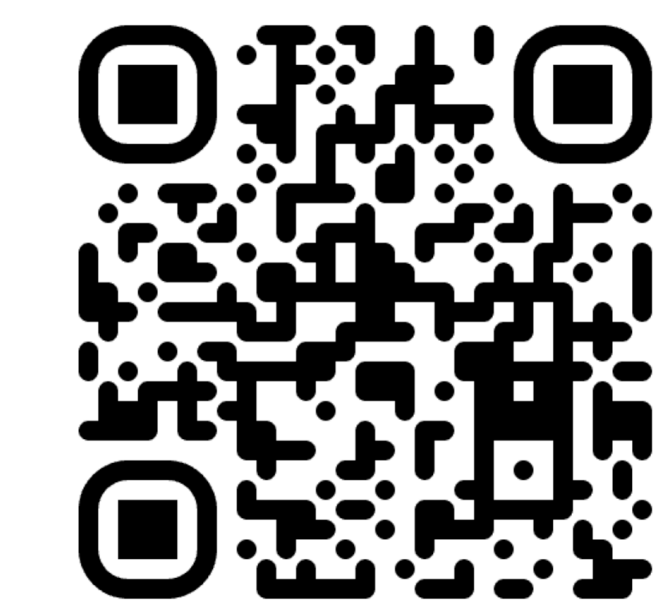




InputDSA: Demixing then Comparing Recurrent and Externally Driven Dynamics

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Highlights

- We introduce **InputDSA**, a method for comparing both **intrinsic** and **input-driven** dynamics of complex systems from data alone.
- InputDSA is **robust to noise, partial observation, and unknown/surrogate inputs**
- We apply InputDSA to **RL-trained RNNs and neural population recordings** from rats.
- InputDSA is the only input-aware dynamics comparison method to date.

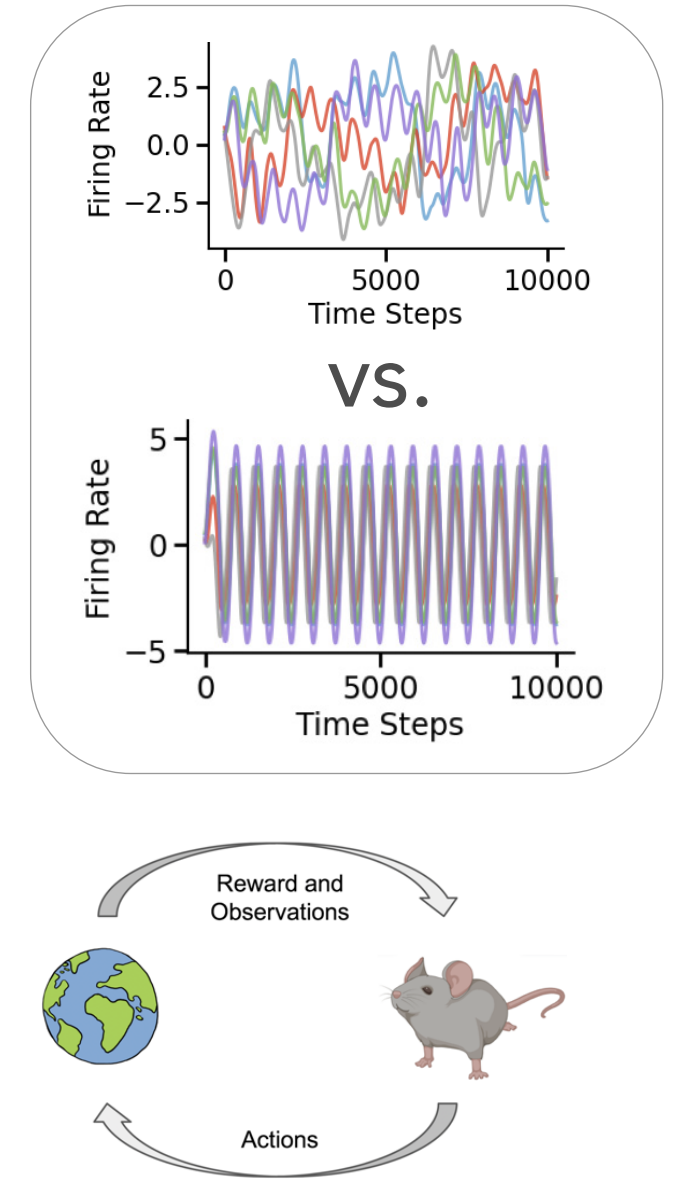
Background

Dynamical Similarity Analysis (DSA)

- Standard comparison methods (CKA, Procrustes, etc.) capture geometry but not dynamics [1,2,3]
- DSA [4] compares dynamical systems by estimating and comparing Koopman Operator approximations via Dynamic Mode Decomposition [5,6]
- But **DSA ignores external inputs**

Koopman operator theory

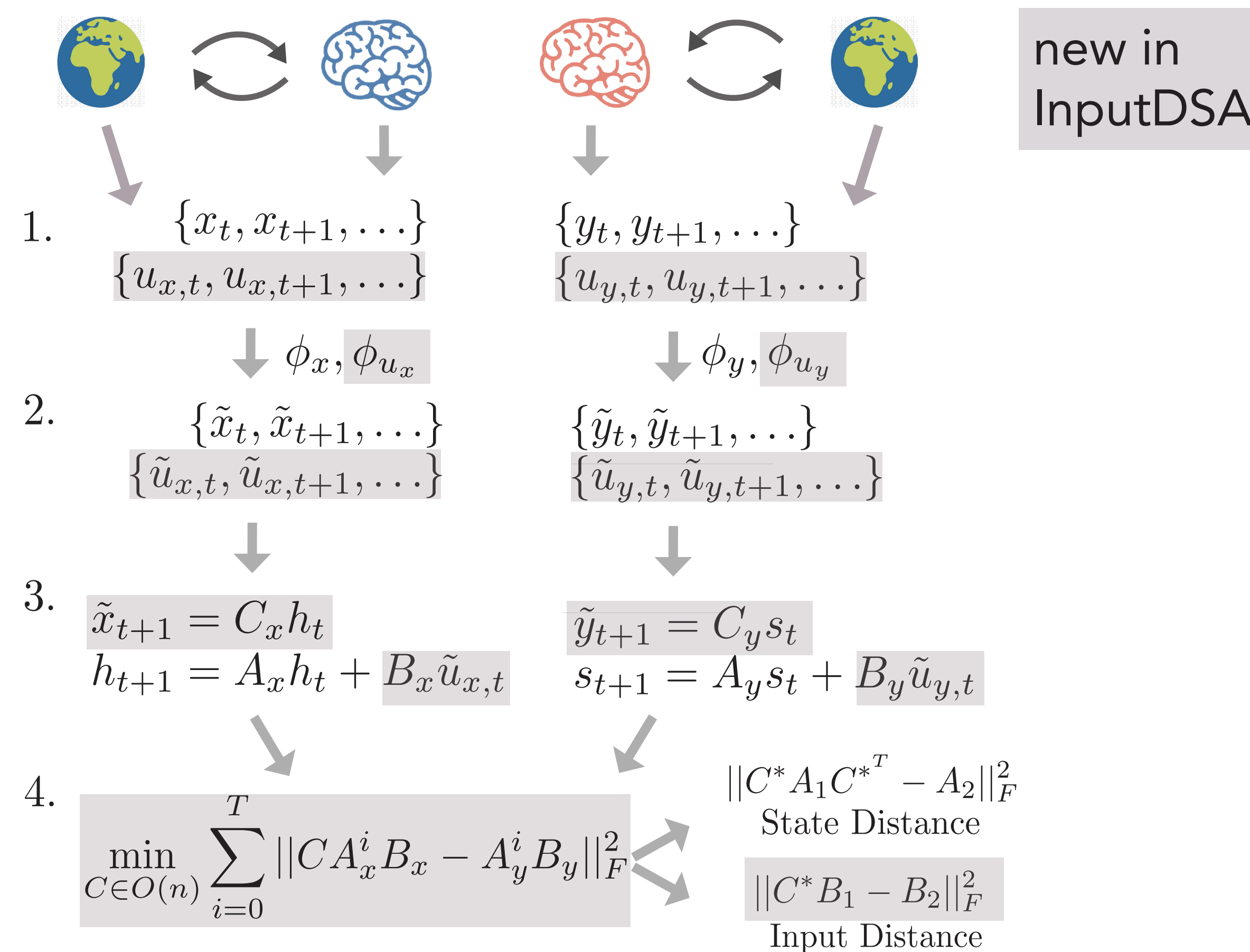
- Nonlinear dynamics $\dot{x} = f(x)$ can be represented as linear evolution in a lifted observation space: $\phi(\dot{x}) = A\phi(x)$ [5]
- Finite approximation via Dynamic Mode Decomposition (DMD): estimate A from data [6]
- With external input: DMD with control (DMDc) estimates both A and B in $\dot{x} = Ax + Bu$ [7]



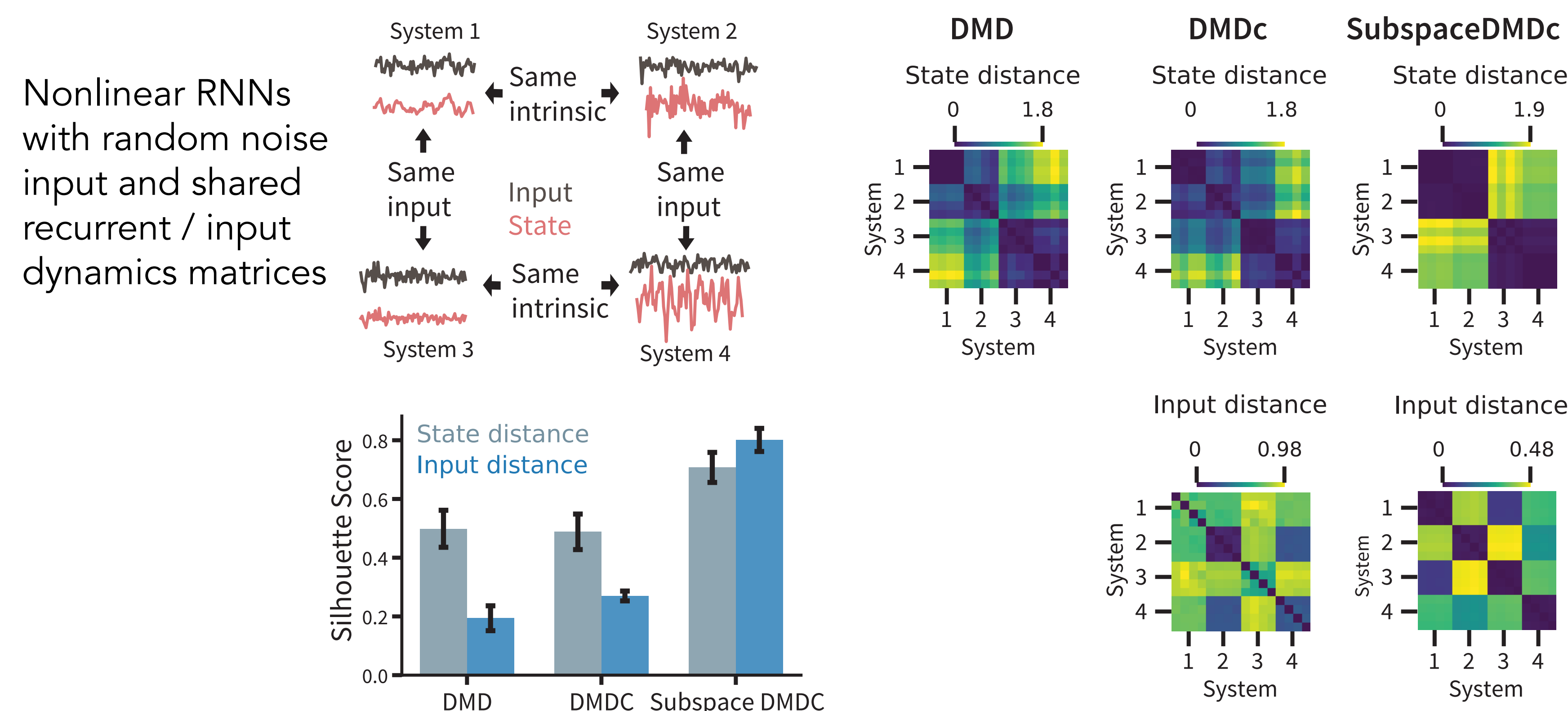
Problem: Existing dynamical comparison methods ignore input drive.

Our Solution: Input-aware Dynamical Similarity Analysis (InputDSA)

1. Sample state and input time series from two dynamical systems
2. Nonlinearly embed these into higher dimensions
3. Fit a linear state space model via SubspaceDMDc
4. Compare recurrent and input dynamics matrices together and separately

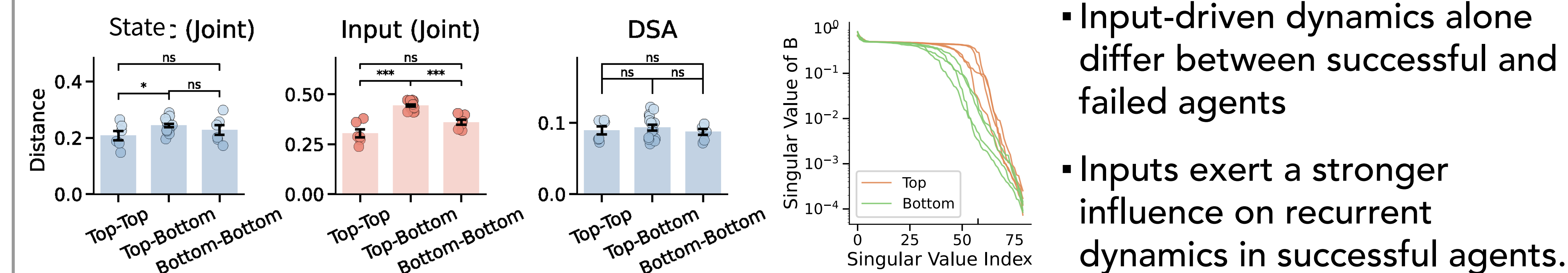
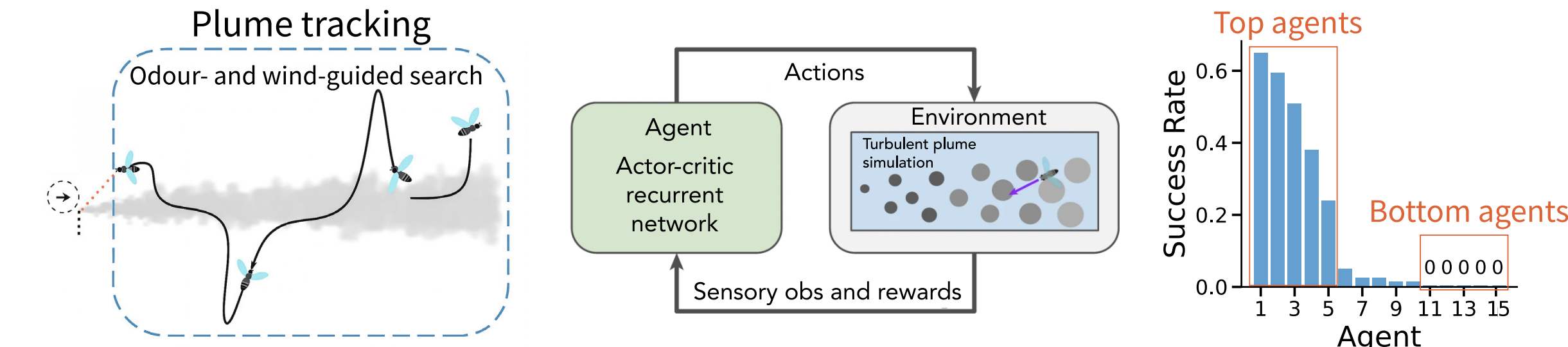


InputDSA disentangles recurrent from input-driven dynamics

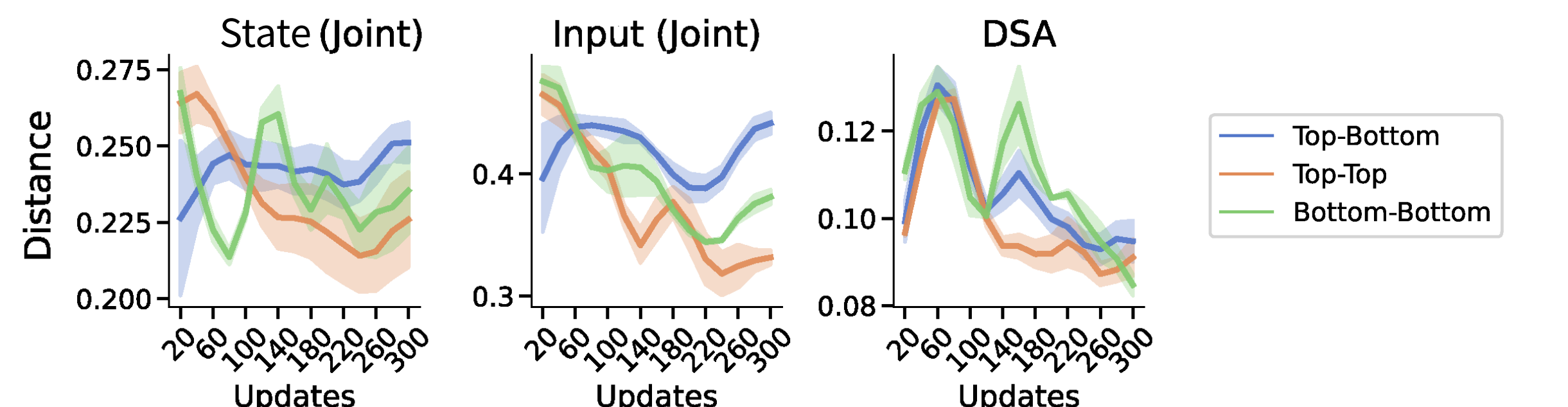


InputDSA tracks the evolution of individual differences over learning

- Deep RL agents (RNNs with soft-actor critic) trained to track the source of an odour plume [9]

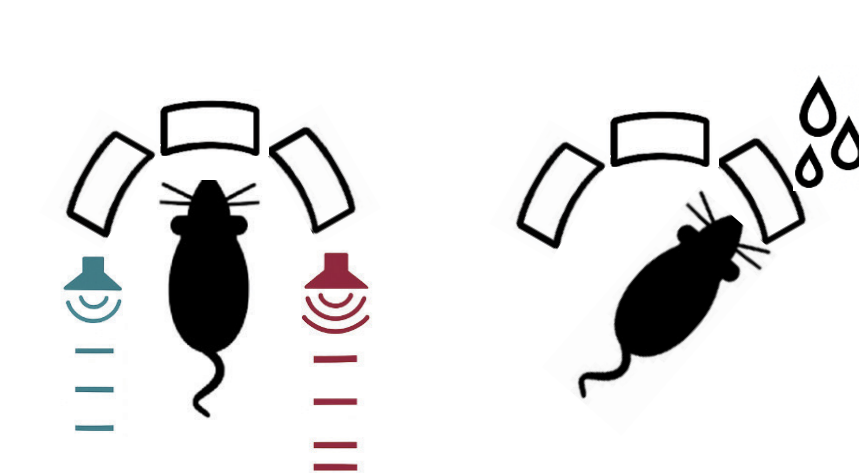


- Effective plume tracking strategies are similar, while worse ones are more varied.

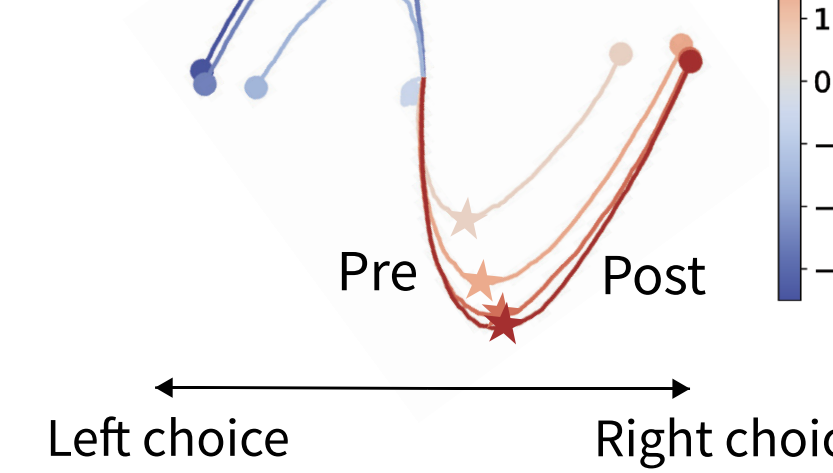


InputDSA captures interpretable differences in neural dynamics over time

Auditory evidence accumulation

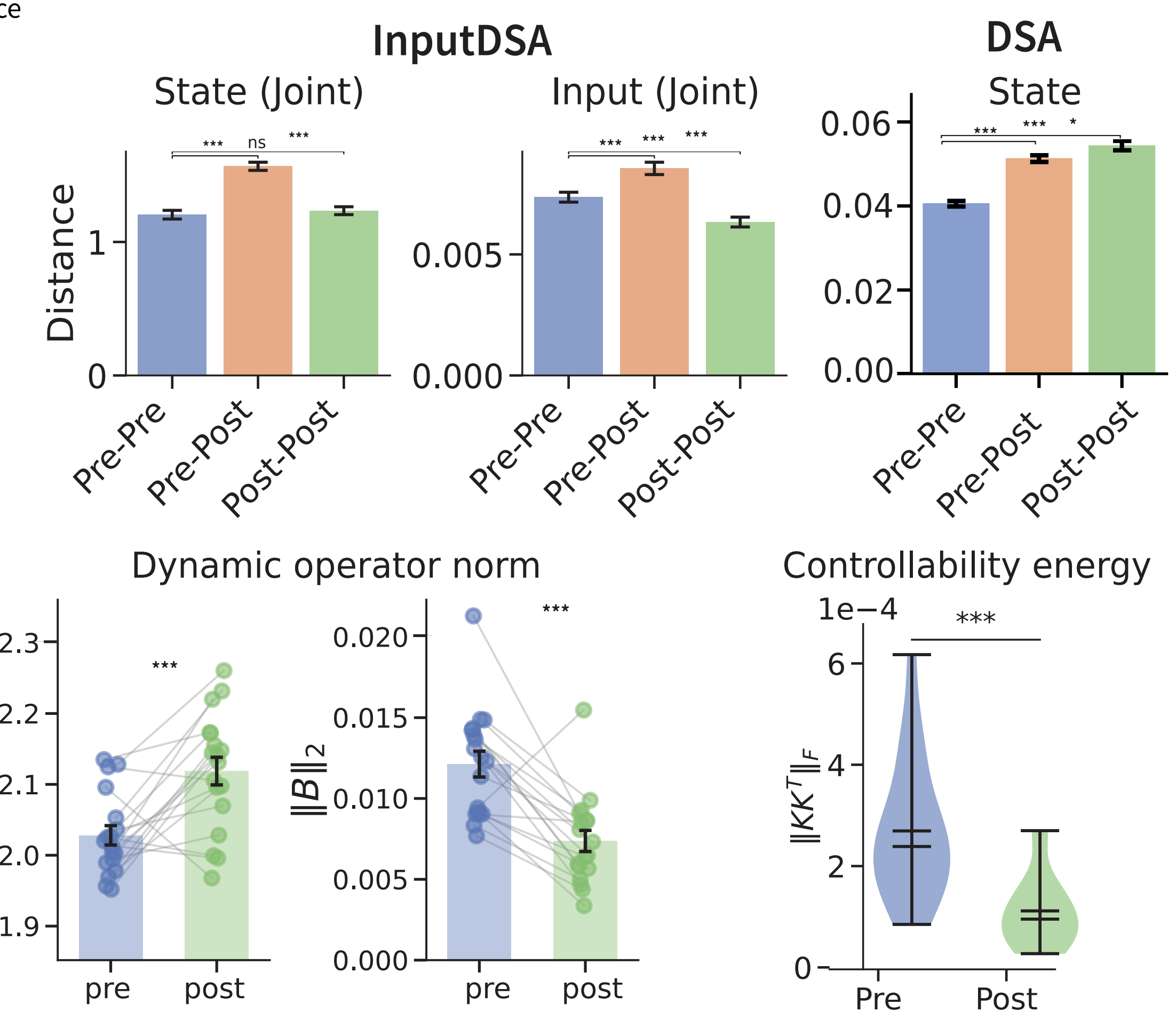


Decision commitment



- High-dimensional neural data recorded from six brain regions in rats performing a sensory integration task [10].
- “Neural time of commitment” (nTc) denotes when the rat has determined their decision.

- InputDSA identifies changes in both input and recurrent dynamics across nTc.
- Input-driven evidence accumulation to more autonomous decision commitment.
- Neural dynamics become less input controllable after time of commitment



Innovations

1. Subspace DMDc for resolving input-driven effect on partially-observed systems [8]
2. Jointly align both input and recurrent dynamics matrices via controllability matrix

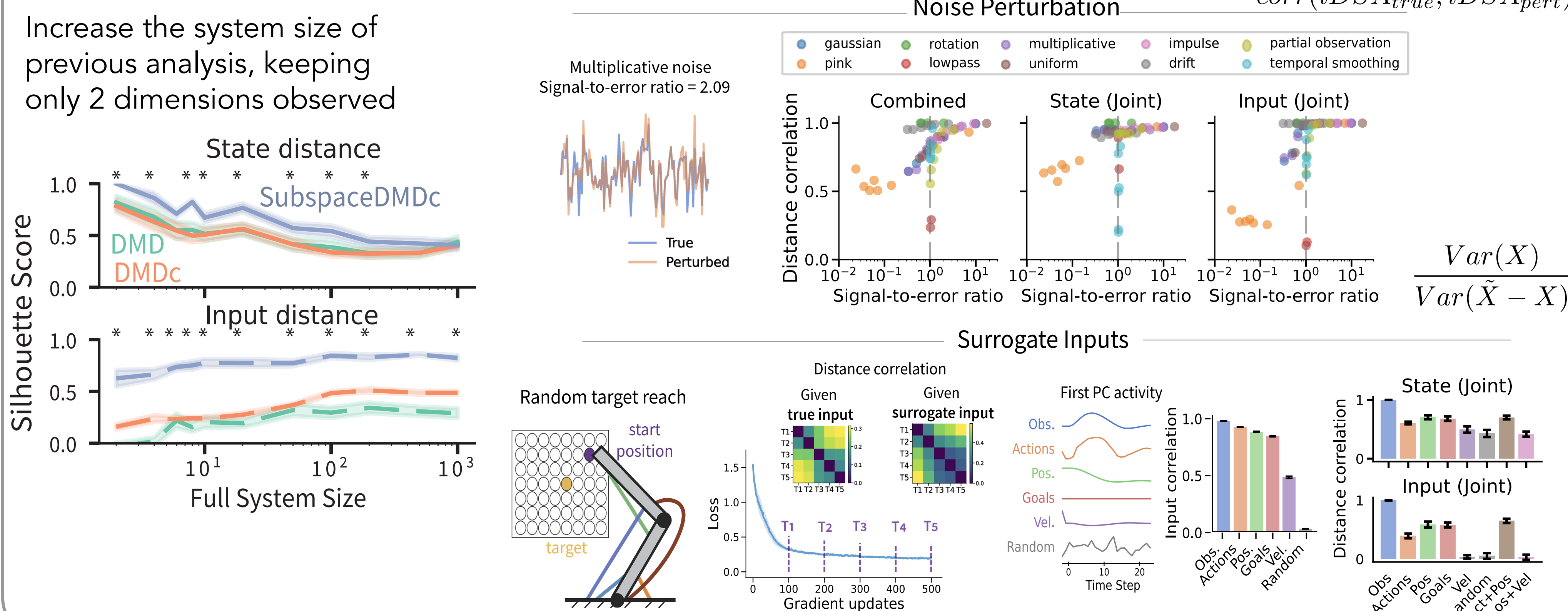
$$K_1(T) = (B_1 \quad A_1 B_1 \quad A_1^2 B_1 \quad \dots \quad A_1^{T-1} B_1)$$

$$y = Cx \implies A_1 = CA_2C^T, B_1 = CB_2 \quad K_1 = CK_2$$

The controllability matrix is only invariant to orthogonal transformations on the state, enabling fast comparisons with Procrustes Analysis:

$$\text{InputDSA}(K_1, K_2) = \min_{C \in O(n)} \|CK_1 - K_2\|^T$$

Robustness to noise, partial observation, and unknown inputs



References: [1] Schrimpf et al 2018, [2] Williams et al., 2021, [3] Kornblith et al. 2019, [4] Ostrow et al. 2023, [5] Koopman 1931, [6] Brunton et al. 2021, [7] Proctor et al. 2014, [8] Verhaene et al. 1994, [9] Singh et al. 2023, [10] Luo et al. 2025